Sequential Causal Training of Windowed PINNs for the Lorenz ODE System

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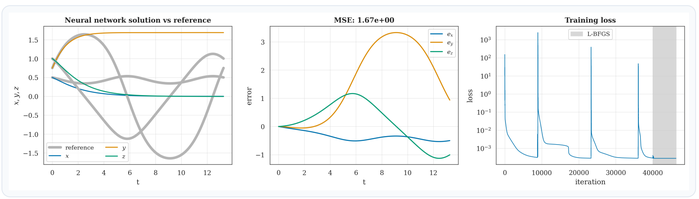
Final Year Design Project (FYDP-2) Defense

Code & saved run

The Long-Horizon Problem

Training one network per time window with causal weights cuts the combined RMSE from 2.237 to 6.97×10^{-5}, using only the ODE residual.

- A full-interval PINN couples all times, yet later states depend on earlier ones.
- Errors pile up and cannot be undone. Our saved single-network run stalls at RMSE 2.237.
- Builds on causal training (Wang et al., 2024).



Single network on the full interval. The trajectory drifts from the DOP853 reference and the loss plateaus (RMSE 2.237).

Lorenz-1960 IVP

GOVERNING EQUATIONS

$$\dot{x} = -0.1\,yz, \qquad \dot{y} = 1.6\,xz, \qquad \dot{z} = -0.75\,xy$$

Initial state u_0	(0.5, 0.75, 1.0)
Coefficients (k = 2, l = 1)	(−0.10, 1.60, −0.75)
Horizon	[0, 13.26446] · nearly closed orbit
Reference	DOP853 · rtol 10^{-10} · atol 10^{-12}
Evaluation	13,265 uniform times
Role of reference	monitoring only, never in the loss

- The reference ends almost where it starts: a **nearly closed orbit**, not an exact period.

Windowed PINN

- Each window k gets its own 4×60 tanh network. Windows share boundaries, not parameters.
- A hard trial solution fixes each initial state **by construction**, so there is no IC penalty to weight.
- Each window starts from the previous predicted endpoint and copies its weights as a warm start.

HARD TRIAL SOLUTION: EXACT IC

$$\hat{u}_k(t) = u_{0,k} + (t - t_k)\, N_k(t;\theta_k)$$

WINDOW HANDOFF + WARM START

$$u_{0,k+1} = \hat{u}_k(t_{k+1}), \qquad \theta_k \leftarrow \theta_{k-1}$$

Run Configuration

Interval / windows	[0, 13.26446] / 27 equal windows
Parameters	11,283 per window · 304,641 total
Collocation	1,536 uniform per window · 41,446 unique points
Optimizer	Adam 10^{-3} rate, ×0.9 every 1,000 steps
Causal stages	ε = 0.01, 0.1, 1, 10 · cap 4,000 steps
L-BFGS	≤ 1,000 closures per window
Precision / seed	float64 / seed 0

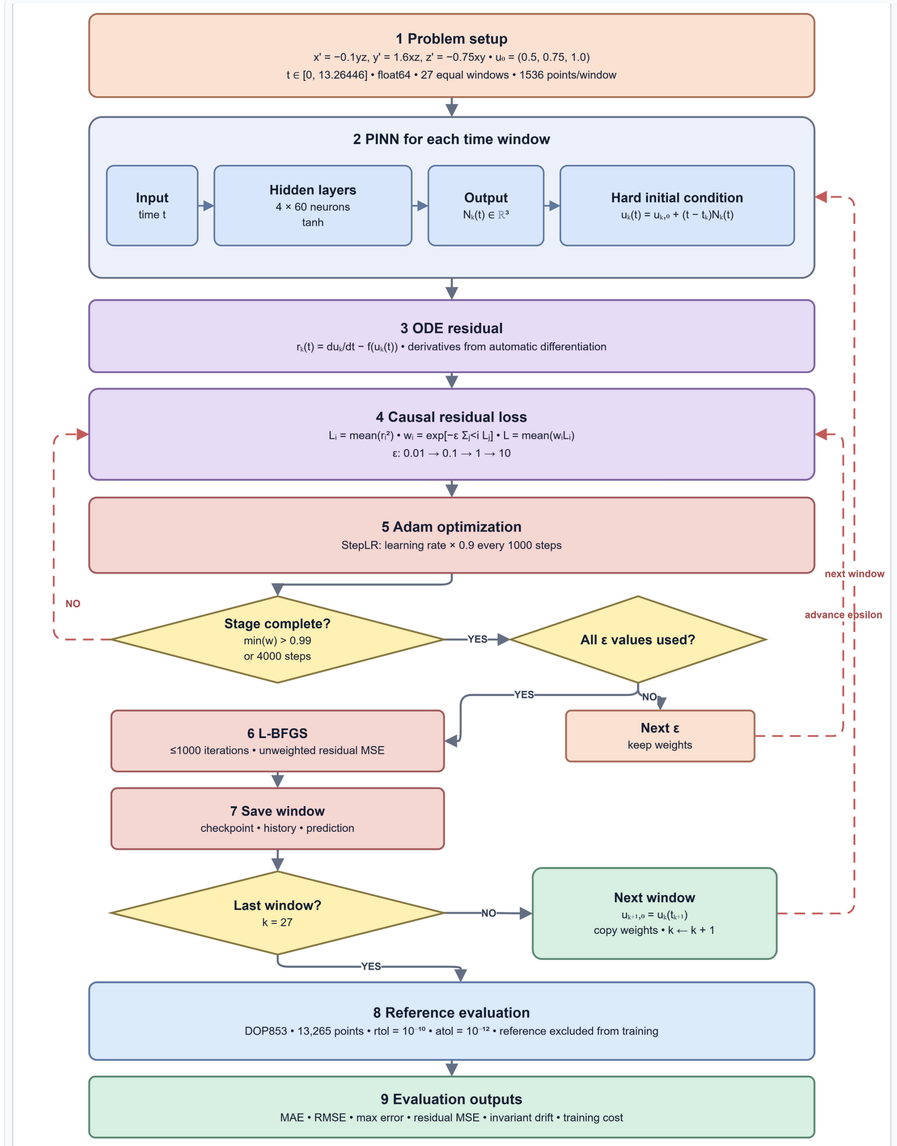
One Network vs 27 Windows

Measure	One network	27 windows
Trainable parameters	11,283	**304,641**
Training points	39,793 LHS	**41,446 uniform**
Combined state RMSE	2.237	**6.97×10^{-5}**
Max state-error norm	3.346	**3.09×10^{-4}**
Collocation residual MSE	2.80×10^{-4}	**1.87×10^{-7}**
Recorded wall time (s)	4,529	**881**

The two runs differ in several options at once, so no single factor is isolated.

34,765 Adam updates · ≤ 1,000 L-BFGS closures · 2 of 108 stages capped

Training Pipeline



Implementation pipeline. Each window runs a hard-IC PINN through four causal ε stages and an L-BFGS polish, then hands off. The DOP853 reference never enters training.

Causal Weighting

- Each point's weight depends on the accumulated loss at all earlier points, so the network fits early times before late ones.
- Weights are recomputed every epoch and held fixed (detached) during the Adam step.

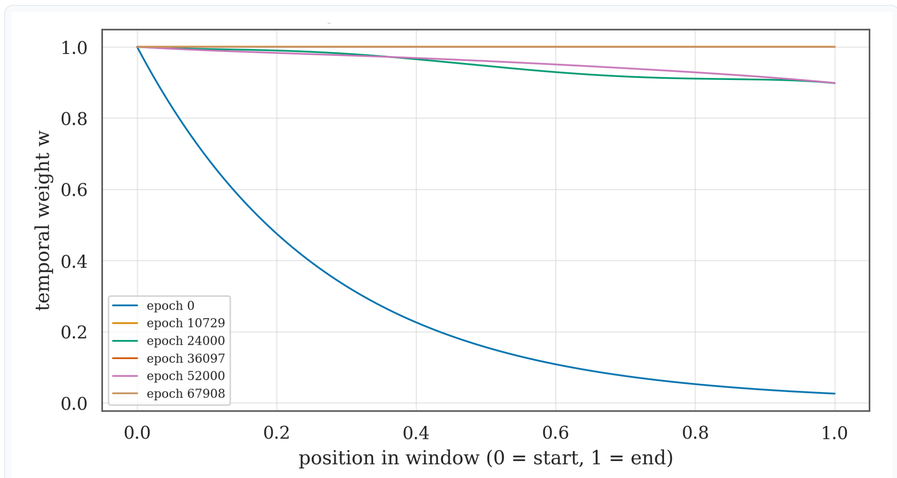
CAUSAL WEIGHT

$$w_{k,i} = \exp\Big(-\varepsilon \sum_{j<i} \mathcal{L}_{k,j}\Big)$$

CAUSAL LOSS

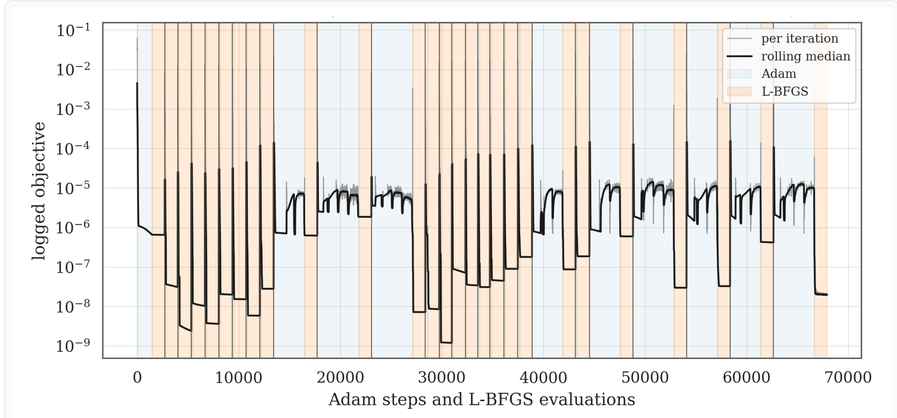
$$\mathcal{L}_{k,\varepsilon} = \frac{1}{n_k} \sum_{i=1}^{n_k} w_{k,i}\, \mathcal{L}_{k,i}$$

advance when min w > 0.99 · then L-BFGS · unweighted residual mean



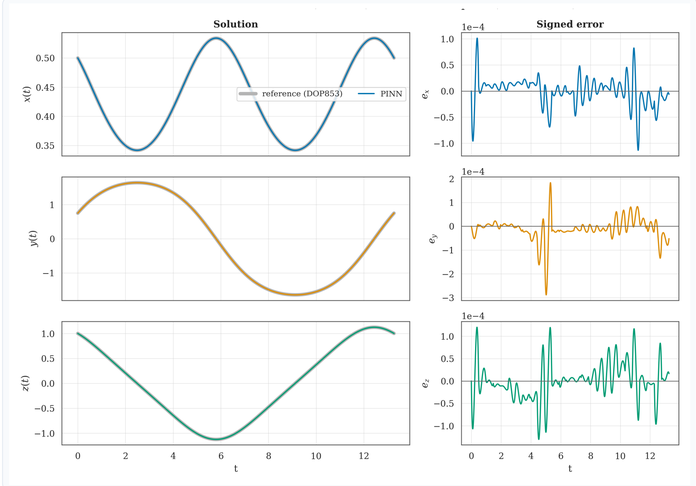
Causal weights at sampled training states. Early on only the window start counts. Weights flatten toward 1 as earlier losses shrink.

Optimizer Sequence



Adam and L-BFGS phases across all 27 windows. Adam minimises the causally weighted loss, L-BFGS the raw residual mean. In every window the objective falls by orders of magnitude before handoff.

Trajectory Accuracy



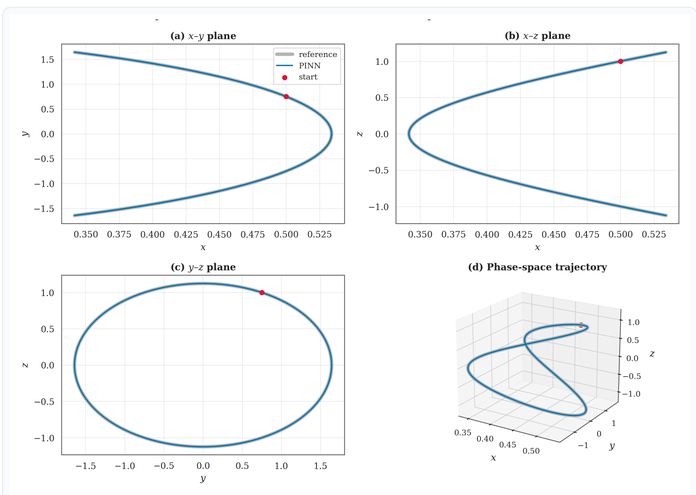
Windowed PINN vs DOP853 on 13,265 uniform times. The curves overlap. Signed errors (×10^{-4}) peak near $t \approx 5$.

State	MAE	RMSE	Max
x	1.89×10^{-5}	2.74×10^{-5}	1.13×10^{-4}
y	2.94×10^{-5}	4.96×10^{-5}	2.88×10^{-4}
z	2.78×10^{-5}	4.05×10^{-5}	1.30×10^{-4}
State norm	**5.12×10^{-5}**	**6.97×10^{-5}**	**3.09×10^{-4}**

Physics Checks

- Residual MSE: 1.874×10^{-7} on collocation points, 1.872×10^{-7} on evaluation points (gap ≈ 1.000).
- State error and ODE residual stay within 10^{-5}–10^{-3} along the whole orbit.
- Values are exactly continuous at all 26 inner interfaces, by construction.
- The DOP853 reference itself drifts by less than 4×10^{-10}.

Phase Portraits



Phase portraits of the learned trajectory. PINN in blue, reference in grey, start u_0 = (0.5, 0.75, 1.0) in red.

Limitations & Future Work

- Each window after the first starts from a **predicted** state, so earlier error carries forward and derivatives can jump at interfaces.
- One seed, one initial state, one coefficient set. No windowed run without causal weights was saved.
- Next: one-change-at-a-time runs over several seeds, and boundary-value problems.

Lorenz (196) Tellus · Raissi, Perdikaris & Karniadakis (2019) J. Comput. Phys. · Lagaris, Likas & Fotiadis (1998) IEEE Trans. Neural Netw. · Wang, Sankaran & Perdikaris (2024) Comput. Methods Appl. Mech. Eng. · Penwarden et al. (2023) J. Comput. Phys. · Matthews & Bihlo (2024) PinnDE · Virtanen et al. (2020) SciPy 1.0

Novelty

A reusable training algorithm for long-horizon ODE initial-value problems.

1. Split the horizon into windows with exact ICs.
2. Train each with staged causal weights + L-BFGS.
3. Hand endpoint and weights to the next window.

SOLVE YOUR OWN SYSTEM: SUPPLY

initial values u_0 · right-hand side $f(u)$ · time span $[0, T]$

github.com/ihmorol/lorenz1960-pinn